

(2) and (3) hold for any $\Gamma \times \mathbb{Z}_2$ -invariant metric on $\hat{G}$. (Such a metric exists since the action of $\Gamma \times \mathbb{Z}_2$ is discrete.) The metric satisfying (4) is constructed in 8.3.F below. Now, let us prove (5). First we observe that there is a $\Gamma \times \mathbb{Z}_2$ -invariant discontinuous quasi-isometry $GX \rightarrow \hat{G}$ sending $\mathbb{R}$ -orbits to $\mathbb{R}$ -orbits. This map can be made continuous with a $\Gamma \times \mathbb{Z}_2$ -invariant partition of unity in GX . What remains is to make such a map $GX \rightarrow \hat{G}$ homeomorphic on every $\mathbb{R}$ -orbit. Let us observe that for every quasi-isometry $f: \mathbb{R} \rightarrow \mathbb{R}$ and for every continuous non-negative function $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ with compact support the φ_2 -diffusion of f defined by

$$f_\lambda(t) = \int_{\mathbb{R}} f(t+\tau)\varphi(\lambda^{-1}\tau)d\tau / \int_{\mathbb{R}} \varphi(\lambda^{-1}\tau)d\tau$$

is a homeomorphism for every sufficiently large $\lambda > 0$, which only depends on the implied quasi-isometry constant of f . This diffusion applies to the map $GX \rightarrow \hat{G}$ on every orbit thus making this map homeomorphic on the orbits. The same construction yields maps $\hat{G}_1 \rightarrow \hat{G}_2$ required by (6).

8.3.E Corollary: *The geodesic flow on a compact manifold V of negative curvature is uniquely determined up to a time reparametrization by $\Gamma = \pi_1(V)$. (In other words the 1-dimensional foliation into orbits is determined by Γ .) Furthermore let A be a finite group of exterior (i.e. in $\text{Aut } \Gamma / \text{Int } \Gamma$) automorphisms of Γ . Then there is a homeomorphic action of A on the unit tangent bundle GV of V sending geodesic orbits to orbits and inducing (in an obvious sense) the given action of A on Γ . (Compare 8.2.)*

Proof: Apply the theorem to the (hyperbolic!) semi-direct product $\Gamma \ltimes A$.

8.3.F Metrics in $\hat{G}$ satisfying (4): Consider a locally compact space H and a partition $\mathcal{L}$ of H into 1-dimensional manifolds called *leaves* (or *orbits*). A *split neighborhood* in H by definition is an open subset in H split into Cartesian product of some metric space by the real line, say $B \times \mathbb{R} \subset H$, such that every fiber $b \times \mathbb{R} \subset \mathbb{R}$, $b \in B$ is contained in a single leaf (depending on b) of $\mathcal{L}$. Call $\mathcal{L}$ a *foliation* if each point in H admits a split neighborhood.

Basic example: The quotient space $\hat{G}/\Gamma \times \mathbb{Z}_2$ partitioned into the images of $\mathbb{R}$ -orbits in $\hat{G}$ is a foliation.

Definition: A *transversal Lipschitz structure* in a foliation $\mathcal{L}$ is given by a metric $\|\cdot\|_B$ in B for each split neighborhood $B \times \mathbb{R} \subset H$, such that for every two neighborhoods $B_1 \times \mathbb{R} \subset B_2 \times \mathbb{R} \subset H$ the obvious map $B_1 \rightarrow B_2$ is a bi-Lipschitz homeomorphism of B_1 onto its image in B_2 .

Basic example: Every metric in $\partial^2 \hat{G}$ for which the action of $\Gamma \times \mathbb{Z}_2$ is Lipschitz induces in an obvious way a transversal Lipschitz structure in the $\mathbb{R}$ -orbits foliation on $H = \hat{G}/\Gamma \times \mathbb{Z}_2$.

Now, let $\mathcal{L}$ be a foliation with a fixed transversal Lipschitz structure. A metric $\|\cdot\|_H$ on H is called *Lipschitz* if for every split neighborhood the identity map

$$(B \times \mathbb{R}, \|\cdot\|_H) \leftrightarrow (B \times \mathbb{R}, \|\cdot\|_B \times \|\cdot\|_{\mathbb{R}})$$

is bi-Lipschitz for the standard metric $\|\cdot\|_{\mathbb{R}}$ in $\mathbb{R}$ and where $\|\cdot\|_B \times \|\cdot\|_{\mathbb{R}}(b,t) = \|b\|_B + |t|_{\mathbb{R}}$.

8.3.G Lemma: Every $\mathcal{L}$ admits a Lipschitz metric. Furthermore such a metric is unique in the following sense. If $\|\cdot\|_1$ and $\|\cdot\|_2$ are two Lipschitz metrics on H then there exists a bi-Lipschitz homeomorphism $(H, \|\cdot\|_1) \leftrightarrow (H, \|\cdot\|_2)$ which sends every leaf of $\mathcal{L}$ into itself and which is arbitrarily close

to the identity map.

Proof: First we prove the uniqueness. The identity map $(H, | \cdot |_1) \rightarrow (H, | \cdot |_2)$ can be approximated by a leaves preserving Lipschitz map $f: (H, | \cdot |_1) \rightarrow (H, | \cdot |_2)$ with a $| \cdot |_1$ -Lipschitz partition of unity in H . Then we apply a φ_λ -diffusion to f with a small (depending on the precision of the preceding approximation) $\lambda > 0$, in order to make f bi-Lipschitz. Routine details are left to the reader.

Now, to prove the existence, decompose H into countable union of split neighborhoods, $H = \bigcup_i B_i \times \mathbb{R}$.

The desired metric on H is obtained by extending such from $H_j = \bigcup_{i \leq j} B_i \times \mathbb{R}$ to H_{j+1} , where the uniqueness of the metric on $(B_{j+1} \times \mathbb{R}) \cap H_j$ ensures the possibility of such an extension. Again, the details are left to the reader.

8.3.H Corollary: Let $| \cdot |$ be a metric on $\partial^2 \hat{G}$ for which the action of $\Gamma \times \mathbb{Z}_2$ is Lipschitz. Then there exists a $\Gamma \times \mathbb{Z}_2$ -invariant metric in $\hat{G} = \partial^2 \hat{G} \times \mathbb{R}$ for which the identity map

$$(\hat{G}, | \cdot |_{\hat{G}}) \mapsto (\partial^2 \hat{G} \times \mathbb{R}, | \cdot | \times | \cdot |_{\mathbb{R}})$$

is bi-Lipschitz.

8.3.I The proof of (4): Now we involve the metric $| \cdot |_\mu$ on $\partial \hat{G}$ defined in 7.2 and observe that the exponential convergence property is "transversally" satisfied for the corresponding "transversally defined metric" on $\hat{G} = \partial^2 \hat{G} \times \mathbb{R}$. Hence, this property is satisfied for the (true) metric provided by 8.3. Q.E.D.

8.3.J Remark: The construction of $\hat{G} = \hat{G}(\Gamma)$ can be performed in a more canonical way without using any auxiliary space X . Namely $\hat{G}$ can be obtained as a quotient of the space $\partial^3 \Gamma$ by using functions $L(a, b, x_1, x_2)$ where now x_1 and x_2 are points in $\partial \Gamma$ different from a and b , where L satisfies (1), (2) and (3) of 8.3.D and

where (4) is replaced by

$$L(a,b;x_1,x_2) \rightarrow \infty$$

for $x_1 \rightarrow a$ and $x_2 \rightarrow b$.

8.3.K Example: If $\partial\Gamma$ consists of two points, then our theorem reduces to the following well known (and nearly obvious) fact.

Let Γ be a finite extension of $\mathbb{Z}$. Then there exists a homomorphism h of Γ into the (only) non-trivial semidirect product $\mathbb{Z} \ltimes \mathbb{Z}_2$, such that the image $h(\Gamma) \subset \mathbb{Z} \ltimes \mathbb{Z}_2$ is infinite.

8.3.L Ends of non-hyperbolic groups: Theorem 8.3.C and its proof resemble Stallings theorem on groups with infinitely many ends. In fact Stallings theorem admits an extremely short geometric proof (essentially due to Matthew Brin) based on the regularity of solutions of Plateau's problem (see [Gr5]). Here is another version of this proof where we only need the regularity of harmonic functions on Riemannian manifolds.

Step 1: To simplify the matter we assume Γ to be finitely presented (the general case can be treated along the lines indicated in [Gr5]) and then take a closed Riemannian manifold V with $\pi_1(V) = \Gamma$ such that the space of ends of Γ , denoted $\partial_0\Gamma$, is identical to that of the universal covering X of V .

Step 2: We assume Γ to have more than one end. Then, either Γ is a finite extension of $\mathbb{Z}$ and $\partial_0\Gamma = \partial\Gamma = \{\pm\infty\}$, or $\partial_0\Gamma$ is a Cantor set. Cover $\partial_0\Gamma$ by two disjoint non-empty open (and hence closed subsets) A and B and consider all smooth functions f on X which continuously extend to $X \cup \partial_0X$ and have $f|_A = 0$ and $f|_B = 1$. Then we consider the Dirichlet functional $\|f\|_p^1 = \int_X \|\text{grad } f\|^p$ on these functions and define

$$|A-B|_p = \inf_f \|f\|_p^1.$$

Clearly $|A-B|_p < \infty$ for all $p > 0$. If $p = 1$, then $|A-B|_p$ equals the $(n-1)$ -dimensional (for $n = \dim X$) volume of a minimal hypersurface $H \subset X$ separating A from B . If $\partial_0 \Gamma = \{\pm\infty\}$ and $p > 1$, then $|A-B|_p = 0$. However, if $\partial_0 X = \partial_0 \Gamma$ is infinite, then

$|A-B|_p > 0$. Moreover $\|\partial_0 X\|_p \stackrel{\text{def}}{=} \inf_{A,B} |A-B|_p > 0$. This follows from the (obvious) exponential lower bound on the number of ends of X detected in the ball of radius $R \rightarrow \infty$ around a fixed point in X . Notice that the group Γ plays little role here: the same conclusion would hold for every X' quasi-isometric to X .

Step 3: If $p \geq 1$ and $|A-B|_p > 0$, then there exists a function f for which $\|f\|_p^1 = |A-B|_p$, though f may be non-smooth for $p \neq 2$. For example if $p = 1$ this extremal f equals zero on the A -side of the hypersurface H and one on the B -side. However, for $p = 2$ this function (obviously) is harmonic and hence smooth. Notice that Γ plays no role here either.

Step 4: There exists a function $\bar{f}$ on $X \cup \partial_0 X$ which is continuous on $\partial_0 X = \partial_0 \Gamma$ with values 0 and 1 and such that $\|\bar{f}\|_p^1 = \|\partial_0 X\|_p$, where $p \geq 1$ and provided $\|\partial_0 X\|_p \neq 0$ (e.g. $\partial_0 \Gamma$ is infinite and $p = 2$). This is obvious since the isometry group of X (which contains Γ) is cocompact. This is the only moment where Γ is seriously used.

Step 5: Notice that for every f and γ the functions

$$\gamma_+ f = \max(f, \gamma f)$$

and

$$\gamma_- f = \min(f, \gamma f)$$

satisfy

$$\|\gamma_+ f\|_p^1 + \|\gamma_- f\|_p^1 = 2\|f\|_p^1.$$

Hence, if $\bar{f}$ is our absolutely minimizing function, then

$$\|\gamma_+ \bar{f}\|_p^1 = \|\gamma_- \bar{f}\|_p^1 = \|\bar{f}\|_p^1.$$

Now, if every extremal $\bar{f}$ is known to be sufficiently regular (C^1 -smooth is enough), we conclude that the function $\bar{f}$ has the following property: for every $\gamma \in \Gamma$ one of the two inequalities holds

$$(1) \quad \bar{f}(x) \geq \bar{f}(\gamma(x)) \text{ for all } x \in X;$$

$$(2) \quad \bar{f}(x) \leq \bar{f}(\gamma(x)) \text{ for all } x \in X.$$

In particular this is satisfied for $p = 2$ if $\partial_0 \Gamma$ is infinite. Now we recall that $\bar{f}$ is continuous on $X \cup \partial_0 X$ and that $\bar{f}|_{\partial_0 X}$ is non-constant. The existence of such an $\bar{f}$ immediately implies that Γ is a non-trivial free or HNN-product amalgamated over a finite subgroup.

8.3.M It is not hard to show that every finitely generated group Γ admits a left invariant metric $| \cdot |$ such that $(\Gamma, | \cdot |)$ is hyperbolic $\partial \Gamma = \partial_0 \Gamma$ and the action of Γ on $\partial^3 \Gamma$ (and hence the canonical action of Γ on $\partial_0^3 \Gamma$) is B-finite (but not necessarily cobounded). For example, if Γ_1 and Γ_2 are infinite groups connected at infinity, we start with some word metric $| \cdot |_i$ in Γ_i for $i = 1, 2$, then take $| \cdot |'_1 = \log(1 + | \cdot |_1)$ and finally take the metric free product $\Gamma = (\Gamma_1 * \Gamma_2, | \cdot |'_1 * | \cdot |'_2)$. A similar construction applies to amalgamated and HNN-products, but it would be more interesting to find a direct (not using Stallings' theorem) construction and then to derive Stallings theorem from a suitable generalization of 8.3.

Notice that the action of each $\Gamma_i \subset \Gamma$, $i = 1, 2$, is parabolic on $\partial_0 \Gamma$. Namely, Γ_i keeps fixed a unique point

$a_i \in \partial_0 \Gamma$, $i = 1, 2$, and $\gamma(a) \rightarrow a_i$ for every $a \in \partial_0 \Gamma$ as $a \in \Gamma_i$ goes to infinity.

The compactification of any Γ by $\partial_0 \Gamma$ suggests a general notion of (partially) hyperbolic boundaries of a non (word) hyperbolic group Γ . In particular one may seek a *maximal* hyperbolic boundary similar to the Furstenberg boundary (which is $\partial \Gamma$ if Γ is word hyperbolic). An approach to this problem is indicated in [F1].

8.3.N Remarks on groups of rank ≥ 1 : Recall that the space $\hat{G}$ is acted by three groups, Γ , $\mathbb{R}$ and $\mathbb{Z}_2$, and the three actions agree in a certain manner. A similar situation arises for groups Γ operating on symmetric spaces of range $p \geq 1$, for example for subgroups in $O(p, q)$ for $1 \leq p \leq q \leq \infty$. In this case one takes the flow space of chambers for $\hat{G}$ and the three groups are Γ , $\mathbb{R}^p$ and W , where W is the Weil group and where $\hat{G}/\mathbb{R}^p$ can be identified with the Furstenberg boundary of Γ , provided Γ is a (finite dimensional) lattice. However, one does not know what corresponds to this picture for more general semihyperbolic groups, such as cocompact isometry groups of convex manifolds. This is not even known for cocompact groups acting on convex domains in *symmetric* spaces of $\mathbb{R}$ -rank ≥ 2 .

8.3.P Some corollaries of 8.3: We see with 8.3.C that word hyperbolic groups Γ are indeed very similar to fundamental groups of convex manifolds. In particular, there is a one to one correspondence between periodic orbits of the $\mathbb{R}$ -flow on $\hat{G}/\Gamma$ and conjugacy classes of non-torsion elements γ in Γ .

8.3.Q According to 2.2 every word hyperbolic group Γ admits a discrete cocompact action on a contractible finite dimensional polyhedron P . Then an obvious partition of unity argument delivers a Lipschitz Γ -equivariant map $\hat{G} \rightarrow P$ which necessarily (and obviously) is a quasi-isometry. The images of $\mathbb{R}$ -orbits give a distinguished (depending on the map $\hat{G} \rightarrow P$) class of quasigeodesics in P , called *$\hat{G}$ -lines*, which serve in most respects better than geodesics in P . Any such $\hat{G}$ -line joins two points at the boundary $\partial P = \partial \Gamma = \partial \hat{G}$.

If ℓ_1, ℓ_2 and ℓ_3 are three $\hat{G}$ -lines between a_1, a_2 and a_3 at ∂P , then the resulting (ideal) $\hat{G}$ -triangle Δ with vertices a_i and sides ℓ_i exponentially narrows at infinity like an ideal geodesic triangle Δ_ϵ in the hyperbolic plane H_ϵ of curvature $-\epsilon^2 < 0$ for some constant $\epsilon = \epsilon(\Gamma) > 0$. Namely, there is a distance decreasing map of the (2-dimensional) triangle Δ_ϵ into P such that the sides of Δ_ϵ go to the sides ℓ_i of Δ . In particular, Δ bounds a disk (that is the image of Δ_ϵ) of area $\leq \pi\epsilon^{-2}$. This property of $\hat{G}$ -triangles immediately implies a similar property of all ideal G -simplices in P , which leads to the following important corollary.

8.3.R Denote by $Q^n = Q^n(\partial\Gamma)$ the n -dimensional polyhedron whose k -simplices for $k = 0, 1, \dots, n+1$ corresponds to all $(k+1)$ -tuples of distinct points in $\partial\Gamma$ and let $Q_0^n = Q^n \setminus \partial\Gamma$, where $\partial\Gamma$ is identified with the 0-skeleton $Q^0 \subset Q^n$.

Theorem: There exists a metric $|| \cdot ||_\epsilon$ on Q_0^n for some $\epsilon = \epsilon(\Gamma, n) > 0$ and a continuous Γ -equivariant map $T: Q^n \rightarrow P \cup \partial P$ with the following three properties.

- (1) Each k -simplex Δ^k in Q^n is isometric to some ideal k -simplex Δ_ϵ^k in the hyperbolic space H_ϵ^k of curvature $-\epsilon^2$.
- (2) The map T is the identity on $Q^0 = \partial\Gamma = \partial P \subset Q^n$ and Q_0^n is sent into P .
- (3) The map T is distance decreasing on Q_0^n (for the metric H_ϵ and a given simplicial metric on P). Furthermore T is uniformly quasi-isometric on every simplex Δ^k in Q^n . That is T is ϵ - λ -isometric for some constants ϵ' and λ depending on Γ and n .

The property (1) needs a comment. An ideal k -simplex Δ in

H_ϵ^k by definition is the convex hull of $k+1$ distinct points (vertices) $a_0, \dots, a_k$ in ∂H_ϵ^k . That is Δ is the minimal convex subset in H_ϵ^k with $\partial\Delta = \{a_0, \dots, a_k\} \subset \partial H_\epsilon^k$. For example, every edge $\Delta^1 \subset H^1 = \mathbb{R}$ is isometric to $\mathbb{R}$. Also all ideal triangles in H^2 are isometric. However, simplices Δ^k for $k \geq 3$ are not mutually isometric. In fact, one can assign in a canonical way a *geodesic tree* $S \subset \Delta^k \subset H^k$ with the following properties.

- (a) The tree S has $\ell \leq k - 1$ vertices $x_1, \dots, x_\ell$ inside Δ^k .
- (b) S is the union of some geodesic segments between x_j and of $k+1$ geodesic rays joining each $a_i \in \partial H^k$ with some x_j .
- (c) Denote by $d(s)$ the distance from $s \in S$ to the nearest vertex x_j and let $B_s \subset H^k$ be the ball around s of radius $10 \exp(-\epsilon d(s))$. Then the union $\bigcup_{s \in S} B_s \subset H^k$ contains the simplex Δ^k . In other words, Δ^k exponentially narrows away from the vertices x_j .

- (d) The volume of Δ^k is bounded by $\pi \epsilon^{-\frac{k}{2}} \exp(-\epsilon d)$ for $d = 1/2 \operatorname{diam}\{x_1, \dots, x_\ell\}$. In fact, the simplex of the maximal volume is the regular one (see [Th]) for which S is the union of $k+1$ rays joining the points a_i with a single point $x \in H^k$.

- (e) The geometry of Δ^k is essentially determined by the (simplicial) geometry of the tree S with the distance defined by lengths of paths in S between pairs of points. Namely the embedding $S \rightarrow \Delta^k$ is a ϵ -isometry where the implied constant only depends on k and ϵ . Furthermore, the geometry of Δ within given ball B of radius R "essentially" depends on $B \cap S$ with an error $\sim \exp(-\epsilon R)$. More generally let B be a bounded convex subset in H^k , let $S_B \subset \Delta^k \cap B$ be the convex hull of the intersection $S \cap \partial B$ and let $d(y)$ for $y \in \Delta^k \cap B$ denote the distance from y to the (topological) boundary ∂B of B . Then

$$\text{dist}(y, S_B) \leq 10 \exp(-\epsilon d(y))$$

for all $y \in \Delta^k \cap B$ where $d(y) \geq 10\epsilon^{-1}$.

Example: If $k = 3$ then the graph (generically, for $\ell = 2$) looks like the one sketched in Figure 9 below.

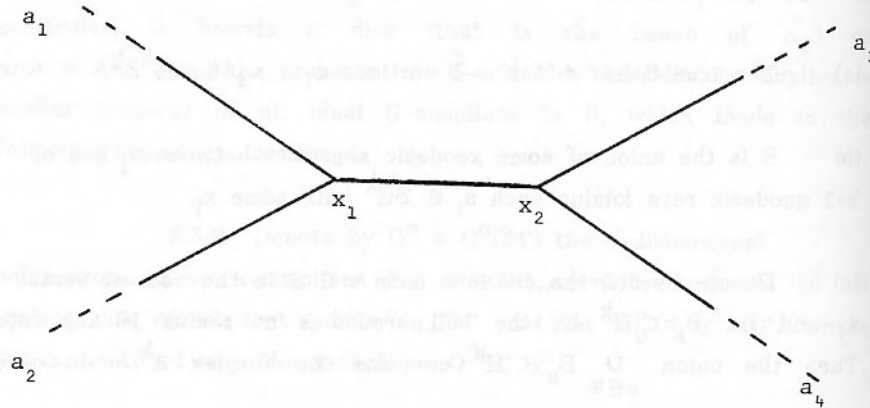


Figure 9

The basic invariant of Δ^3 is the distance $|x_1 - x_2|$. If $|x_1 - x_2|$ keeps bounded then the geometry of Δ^3 is close to that of the regular ideal simplex. If $|x_1 - x_2| \rightarrow \infty$ then Δ^3 (exponentially fast) converges to the union of two triangles spanned by the triples $(a_1, a_2, 1/2(x_1 + x_2))$ and $(a_3, a_4, 1/2(x_1 + x_2))$ correspondingly.

These properties of hyperbolic simplices play a crucial role in Thurston's approach to discrete groups isometrically acting on H^3 and Theorem 8.3 allows us to extend his methods to all word hyperbolic groups. Let us see how it works.

8.3.S Application of geodesic simplices: Let us start with 2-simplices in Γ and to simplify the matter we assume Γ has no torsion. Thus Γ is the fundamental group of a compact aspherical manifold V with boundary and the orbits of the geodesic flow in $\hat{G}(\Gamma)$, mapped to V , are called here *geodesics in V* . We also use some auxiliary Riemannian metric in V to measure length and area but "geodesics" refer to those coming from $\hat{G}$.

Theorem: Let F_0 be a compact connected surface mapped to V . Then there is a homotopy of F_0 to a surface F_1 , such that this homotopy does not move the boundary $\partial F_0 \rightarrow V$ and

$$\text{Area } F_1 \leq C(\text{length}(\partial F_0 = \partial F_1) + |\chi(F_0)|),$$

where χ denotes the Euler characteristic and where the constant $C \geq 0$ only depend on V .

Proof: The surface F_1 is built of two pieces. First is a collar of $\partial F_0 = \partial F_1$ which is obtained by homotopying each component of ∂F_0 (which may be assumed non-contractible) to the unique closed geodesic in the respective homotopy class. This collar can be chosen, such that

$$\text{Area Collar} \leq C \text{ length } \partial F_0,$$

according to 7.4.

Now we have a surface with geodesic boundary ∂' and we homotope it without moving the boundary to a surface with area $\leq C|\chi(F_0)|$. To do this we subdivide F_0 into ideal triangles (compare [Th]) Δ_i for $i = 1, \dots, k = -2\chi(F_0)$ whose sides, in the universal covering of F_0 , are asymptotic to geodesics covering boundary components. Then we take corresponding geodesics in V and thus fill in ∂' by k our triangles. Q.E.D.

Remark: The above construction gives a bound on the total geometry of F_1 , not only on area F_1 . Namely, there exists a constant $\epsilon = \epsilon(V) > 0$, and a metric g of constant curvature $-\epsilon$ on F_1 for which the map into V is distance decreasing and each component S of $\partial F_0 \hookrightarrow V$ has

$$\text{length}_g S \leq C \text{ length}_V S.$$

Moreover, if some S is a closed geodesic in V then S is geodesic in (F_1, g) . Thus we have a perfect replacement of minimal surfaces in manifolds with $K < 0$.

8.3.T Another standard application of the boundness of volumes of our simplices is the boundness of the cohomology $H^{k \geq 2}(\Gamma; \mathbb{R})$ (compare [Gr2]). Since we only have *ideal* simplices the discussion in [Gr2] needs a minor readjustment. (A judicious reader may notice a little problem for $k = 2$ which disappears on second thoughts.)

Remark: The above applications of the geodesic flow can be generalized to most (non-word) hyperbolic groups, by extracting and generalizing the essential information recorded in the flow. Unfortunately, precise statements and proofs become quite cumbersome. (Of course, there is no problem for groups operating on convex spaces with $K \leq -\epsilon < 0$.)

8.4 Symbolic dynamics for word hyperbolic groups: Consider a graph X with the vertex set $X^0 \subset X$ and let Γ be a semigroup of simplicial embedding $X \rightarrow X$. Take a finite set S and let Σ be the space of maps $\sigma: X^0 \rightarrow S$ with the following topology: a sequence σ_i converges to σ for $i \rightarrow \infty$ iff $\sigma_i(x) = \sigma_{i_0}(x)$ for every given $x \in X^0$ and all $i \geq i_0$ for some $i_0 = i_0(x)$. The space Σ with the induced action by Γ is called the *Bernoulli shift* defined by X , Γ and S .

Examples: Let Γ be a group with a fixed generating set $G \subset \Gamma$. Then Γ acts on the corresponding Cayley graph and we have the associated Bernoulli shift on $\Sigma = \Sigma(\Gamma, G, S)$. If Γ is the free cyclic group generated by a single element then our definition agrees with the classical notion of *the two sided Bernoulli shift*. Recall that *one sided shifts* correspond to the action of $\mathbb{Z}_+$ on the obvious graph $\approx \mathbb{R}_+ \supset \mathbb{Z}_+$.

8.4.A Markov shifts: A Γ -invariant subset $\Phi \subset \Sigma$ is called *Markov* if there exists a number $R < \infty$ and a subset $\Sigma(R, x)$ in the set of map $B(x, R) \rightarrow S$, for all $x \in X^0$, where B denotes the R -ball around x , such that a function $\sigma \in \Sigma$ is contained in Φ if and only if $\sigma|_{B(x, R)} \in \Sigma(R, x)$ for all $x \in X^0$.

In most cases of interest the graph X is locally finite and the action of Γ is cofinite; that is there are only finitely many Γ -orbits of points $x \in X^0$. Such shifts are called of *finite type* as they are described by finite combinatorial data given by (finitely many of finite) subsets $\Sigma(R, \bar{x})$ corresponding to Γ -orbits $\bar{x}$ in X^0 .

Example: The diagonal action of Γ on $\Sigma \times \Sigma$ is Markov for the obvious embedding

$$\Sigma \times \Sigma \subset \Sigma(X, \Gamma, S \times S).$$

Also the diagonal $\Delta \subset \Sigma \times \Sigma$ is Markov. Furthermore, if $\Phi \subset \Sigma$ is Markov or of finite type then $\Phi \times \Phi$ and the diagonal in $\Phi \times \Phi$ also are such.

8.4.B Finitely presented systems: Let X and Γ be as earlier and let Ω be a topological space acted upon by Γ . Call Ω *finitely presented* over X if there exists a subshift of finite type in some Bernoulli shift Σ and a surjective Γ -equivariant map $\pi: \Phi \rightarrow \Omega$ such that the equivalence relation $R_\pi \subset \Phi \times \Phi$ defined by

$$(\sigma_1, \sigma_2) \in R_\pi \Leftrightarrow \pi(\sigma_1) = \pi(\sigma_2),$$

is Markov and the topology in Ω equals that in the quotient space Φ/R_π .

3.4.C Theorem: Let X be a hyperbolic locally finite graph and let Γ be a cobounded automorphism (isometry) group of X . Then the induced action of Γ on ∂X is finitely presented. Moreover there is a

Markov presentation $\pi: \Phi \rightarrow \partial X$ which is a finite-to-one map. In particular the action of every word hyperbolic group Γ on $\partial \Gamma$ admits such a presentation over the Cayley graph of Γ .

Proof: The "coding" π_0 defined in 7.6 clearly does the job.

8.5 Markov trees: Let T be a locally finite simplicial tree with our usual metric where the edges have unit length. Fix a reference vertex $x_0 \in T$ and assign to each vertex $x \in T$ the subtree $T_x \subset T$ consisting of those $y \in T$ for which the segment $[x_0, y]$ contains x . Call T Markov or of finite type if there are at most finitely many isomorphism classes of pointed trees (T_x, x) . Label these classes by indices $0, 1, \dots, i, \dots, k$ (where 0 corresponds to the class of (T, x_0)) and define a Markov chain by the matrix $M = M(i, j)$ for $0 \leq i, j \leq k$ where $M(i, j)$ is the number of j -vertices following an i -vertex. Here x is called an i -vertex if T_x is labeled by i and y follows x if $|x - y| = 1$ and $|x_0 - y| = |x_0 - x| + 1$. Now one can easily count the number $N(r)$ of vertices in the sphere of radius r around x_0 for $r = 1, 2, \dots$ as well as the number $N_i(r)$ of i -vertices in this sphere. Namely, $N_i(r) = M^r(0, i)$ and $N(r) = \sum_i N_i(r)$

where M^r is the r -th (matrix) power of M . In particular one sees that the counting (generating) functions $\sum_r N_i(r)t^r$ and $\sum_r N(r)t^r$ are rational.

8.5.A Markov Γ -trees: Suppose, for some pairs of vertices x and y in T , we are given a set $\Gamma_{x,y}$ of isomorphisms $T_x \rightarrow T_y$ such that

- (1) $\text{Id} \in \Gamma_{x,x}$ for all vertices x ,
- (2) $\Gamma_{yx} = \Gamma_{xy}$,
- (3) $\Gamma_{xz} \supset \Gamma_{xy} \circ \Gamma_{yz}$.

Then we call such a Γ -tree *Markov* if there is at most finitely many of Γ -isomorphism classes of trees T_x . Again, we label these Γ -classes by $0, 1, \dots, i, \dots, k$, assign a Markov chain $M(i, j)$ to (T, Γ) and conclude to the *rationality of the corresponding counting function* $\sum_r N_i(r) t^r$.

8.5.B Let X be a locally finite graph with a reference point x_0 and the usual simplicial metric and let T be the set of all minimal segments of integer length issuing from x_0 . This T has a natural structure of a tree, such that the sphere of radius r in T around x_0 consists of segments of length r (joining x_0 with the points of the r -sphere in X).

8.5.B₁ Theorem: *If X is hyperbolic and the isometry group Γ of X is cofinite on X , then the tree T of minimal segments in X issuing from x_0 is Markov. Furthermore if Γ_0 is an isometry group of X having an invariant quasiconvex subset $X_0 \subset X$ on which the action of Γ_0 is cofinite, then the corresponding tree T for the quotient graph X/Γ_0 also is Markov.*

Proof: Take two vertices x and y in X and suppose there exists an isometry $\gamma: X \rightarrow X$, such that $\gamma(x) = y$ and such that the function $|x| - |\gamma(x)|$ is constant on the ball of radius d around x . If $d \geq 100\delta$ (for the simplicial hyperbolicity constant δ of X) then for any two segments $[x_0, x]$ and $[x_0, y]$ the corresponding subtree $T_x \subset T$ is sent onto T_y by γ and the proof follows from the (obvious) finiteness of the number of pertinent (distance) functions/const on d -balls in X .

8.5.C Let X be the Cayley graph of a group Γ with fixed generators $\gamma_1, \gamma_2 = \gamma_1^{-1}, \gamma_3, \gamma_4 = \gamma_3^{-1}, \dots, \gamma_{2n} = \gamma_{2n-1}^{-1}$. Then every oriented edge e of X is labeled by some generator γ_i and the change of the orientation switches γ_i to γ_i^{-1} . Now, the ordering of the generators gives an order to the edges issuing from a fixed vertex $x \in X$ which induces a lexicographic ordering on the set of

segments issuing from x . In particular, there is an order on the set of segments from x to a fixed point x_0 . Let $[\overleftarrow{x_0}, x]$ be the first segment for this order. The set of first segments obviously form a tree, say T , which is embedded into X by $[\overleftarrow{x_0}, x] \mapsto x$. Clearly this embedding is bijective on the set of vertices and preserves the distance to the reference point.

8.5.D If Γ is word hyperbolic then the tree T is Markov (compare [Can]). In particular, the counting function for the number of points in the sphere of radius r in Γ is rational (compare 5.2). Furthermore, if Γ_0 is a quasiconvex subgroup in Γ then the counting function for the number of Γ_0 -orbits in Γ (here we consider the left action of Γ_0 which is isometric for the word metric in Γ) within distance r from the identity element in Γ also is rational. (Notice that every cyclic subgroup in Γ is quasiconvex.)

The proof is identical to that of 8.5.B₁.

8.5.E Observe that the full isometry group Γ of a locally finite graph X is locally compact and totally disconnected but not necessarily discrete. If X is hyperbolic, the action of Γ is cofinite on X and if Γ admits a discrete cocompact (and hence word hyperbolic) subgroup Γ' then the above theorem implies the rationality for the counting function of the number of concentric spheres in X . In fact, one can do without Γ' by using the tree $T(X)$ constructed in 7.6 instead of T . Indeed, this tree is Γ -Markov by the previous argument and the set of vertices in $T(X)$ corresponding to one-point subsets in X clearly is Γ -invariant. Hence, the previous argument applies and leads to the following

8.5.F Theorem: Let X be a locally finite hyperbolic graph whose full isometry (i.e. automorphism) group is cofinite on X (i.e. has at most finitely many orbits on the vertex set of X) and let

X_0 be a quasiconvex subgraph (i.e. there exists a constant $\epsilon > 0$ such the distance minimizing segment in X between any two points in X_0 is ϵ -close to X_0), such that the isometry group Γ_0 of the pair (X, X_0) is cofinite on X_0 . Then the counting function for the number of points on concentric spheres $S(r) \subset X/\Gamma$, $r = 1, 2, \dots$, around a fixed point in the quotient graph X/Γ is rational. In particular, the counting function for the spheres in X is rational. Moreover the counting function for the intersections of concentric r -spheres in X with X_0 also is rational.

8.5.G Remark: If X is an arbitrary geodesic hyperbolic space, $X_0 \subset X$ is a quasiconvex subspace and Γ is an isometry group of the pair (X, X_0) whose action is cobounded on X_0 , then, obviously, the quotient space X/Γ is hyperbolic.

8.5.H Markov spaces: Consider a Markov action of $\mathbb{Z}_+$ on some (necessarily compact) space Φ of functions $\varphi(i)$, $i \in \mathbb{Z}_+$, and call a (non-invariant!) subset $\Psi \subset \Phi$ *semi-Markov* if it is the intersection of a (invariant) Markov subset (subshift) and a *basic* subset $\Psi_0 \subset \Phi$. Recall that, by definition, every basic subset is determined by finitely many (finite) subsets $S_i \subset S$, $i = 1, \dots, k$, by

$$\Psi_0 = \{ \varphi(i) \in \Phi \mid \varphi(i) \in S_i \text{ for } i \leq k \}.$$

A compact topological space Ω is called *semi-Markov* if it admits a presentation $\Omega = \Psi/R$ for some semi-Markov equivalence relation $R \subset \Psi \times \Psi \subset \Phi \times \Phi$, where Ψ is semi-Markov in (the space of) a Markov shift Φ .

Semi-Markov spaces can be equivalently described in terms of Γ -trees T with an additional *neighborhood* structure, that is a subset $\mathcal{N}$ in the set of pairs of vertices x and y in T , such that

$$(i) \quad (x, y) \in \mathcal{N} \Rightarrow |x| = |y|,$$

(ii) the (induced) subset $\bar{\mathcal{N}} \subset \partial T \times \partial T$ consisting of pairs of rays $r_1, r_2: \mathbb{R}_+ \rightarrow T$ with $r_1(0) = r_2(0) = x_0$ and having $(r_1(i), r_2(i)) \in \mathcal{N}$ for $i = 1, 2, \dots$, is an equivalence relation on ∂T .

8.5.I Lemma: *Let $\mathcal{N}$ be Γ -invariant and let there be at most finitely many Γ -classes of subtrees $T_x \subset T$ for all x . If the following condition (A) is satisfied then the space is semi-Markov.*

(A) *If two different vertices y_1 and y_2 follow the same vertex x (i.e. y_1 and y_2 lie in T_x and have $|y_1| = |y_2| = |x| + 1$) then the trees T_{y_1} and T_{y_2} are not Γ -equivalent.*

Proof: Let S be the set of Γ -equivalence classes of subtrees T_x where we can assume the class of T_{x_0} to consist only of T_{x_0} and let Φ consist of the maps $\varphi: \mathbb{Z}_+ \rightarrow S$, such that for every $i = 0, 1, \dots$ every vertex x in the Γ -class $\varphi(i)$ is followed by a (necessarily unique by (A)) vertex y in the Γ -class $\varphi(i+1)$. Clearly, ∂T embeds onto a semi-Markov subset in Φ and $\bar{\mathcal{N}} \subset \partial S \times \partial S$ becomes semi-Markov in $\Phi \times \Phi$. Q.E.D.

8.5.J Let us return to our hyperbolic graph X with a cofinite isometry group Γ . For any of the three trees T , $\bar{T}$ and $T(X)$ associated to X one defines a neighborhood relation N as follows. To be specific, we do it here for the tree T of all segments $[x_0, x]$ from x_0 by

$$(x_1, x_2) \in N \Leftrightarrow |x_1| = |x_2| \text{ and } |x_1 - x_2| \leq 10\delta.$$

Recall that the pertinent Γ -structure on T (as well as on $\bar{T}$ and $T(X)$) is given by isometries in T moving $x \mapsto y$ and preserving the function $|x| \pm \text{const}$ on a ball of radius d around x . Now, if d is sufficiently large then (A) clearly is satisfied in case Γ is discrete (hence, word hyperbolic) and X equals a Cayley graph of Γ . Therefore we have

8.5.K Proposition: The boundary $\partial\Gamma$ of every word hyperbolic group is semi-Markov. Moreover the semi-Markov presentation $\partial\Gamma \rightarrow \partial\Gamma$ and $\partial T(X) \rightarrow \partial\Gamma$ are finite to one.

8.5.L Remark: It is unclear whether ∂X is semi-Markov if the full isometry group Γ of X is non-discrete. (Of course this follows from 8.5 if Γ admits a cocompact lattice.) Notice that there are only countably many (homeomorphism classes of) semi-Markov spaces and so most compact spaces are not semi-Markov. (But, obviously, all finite polyhedra are semi-Markov.) In fact, one does not know if there are only countably many homeomorphism classes of ∂X for the above spaces X and (or) whether there are only countably many of isomorphism classes of (compact) isotropy subgroups $\Gamma_x \subset \Gamma$ for the full isometry group Γ of X . A similar question also is open for locally finite simply connected non-hyperbolic 2-polyhedra X whose full isomorphism group Γ has finitely many orbits on the vertices of X .

8.5.M Let Γ be a word hyperbolic group with a fixed word metric. Take $\gamma \in \Gamma$ and consider the following four subsets,

- (1) The centralizer $C_\gamma \subset \Gamma$. Notice that if γ has infinite order then C_γ is a finite extension of the cyclic group generated by γ .
- (2) The fixed point set $F_\gamma \subset \partial\Gamma$.
- (3) The level

$$D_\gamma(r) = \{x \in \Gamma \mid d_\gamma(x) \leq r\}$$

for $d_\gamma x = |x - \gamma(x)|$ and for some $r \geq |\gamma|$.

- (4) The quasiconvex hull $L_\gamma = L(F_\gamma) \subset \Gamma$ (see 7.5). To be

precise, we take L_γ in the Cayley graph in Γ and then intersect it with Γ .

Observe the following obvious

Lemma: *The above sets satisfy*

- (a) $C_\gamma \subset D_\gamma(r)$
- (b) $D_\gamma(r)$ is invariant under C_γ and the action of C_γ on $D_\gamma(r)$ is cofinite.
- (c) $\partial C_\gamma = \partial D_\gamma(r) = F_\gamma$.
- (d) $D_\gamma(r)$ is contained in some ϵ -neighborhood of L_γ where ϵ depends on γ and r .

Corollaries: It follows from (a) and (b) that the subgroup C_γ is quasiconvex (in the Cayley graph of Γ). Therefore C_γ is finitely generated and word hyperbolic.

According to (c) and 8.1 there are the following four possibilities for $F_\gamma \subset \partial\Gamma$

- (i) F_γ is empty,
- (ii) F_γ consists of two points,
- (iii) F_γ is a nowhere dense, uncountable closed subset in $\partial\Gamma$,
- (iv) $F_\gamma = \partial\Gamma$. In this case Γ/C_γ is finite.

8.5.N Observe that the (displacement) function d_γ on Γ is invariant under the (left) action of C_γ and thus defines a function, say $\bar{d}_\gamma$, on Γ/C_γ such that each level

$$\bar{D}_\gamma(r) = \{x \in \Gamma/C_\gamma \mid \bar{d}_\gamma(x) \leq r\}$$

is finite.

Theorem: The counting function $\sum_r N_\gamma(r) t^r$ for

$$N_\gamma = \#(\bar{D}_\gamma(r+1) \setminus \bar{D}_\gamma(r)), \quad r = 0, 1, \dots,$$

is rational.

Proof: Let us construct the pertinent Markov tree. First, for every finite subset $V \subset \Gamma$ consider the set all segments $[v, \gamma(w)]$ for v and w in V . Let ℓ be the maximum of the lengths of these segments and let $[v_i, \gamma(w_j)]$ be the segments of length ℓ . Now, move these v_i toward w_j by one. That is take the set V' of the points of the form $\ell^{-1}v_i + \ell^{-1}(\ell-1)\gamma(w_j)$. Then remove from V all points v_i (having $|v_i - \gamma(w_j)| = \ell$ for some $w_j \in V$), add V' to what is left and call the result $P_\gamma(V)$. Finally, apply the construction in 7.6 to P_γ instead of P and to subsets $V \subset \Gamma$ of diameter $\leq D$ for $D = 100(|\gamma| + \delta + 1)$. Since P_γ is C_γ -invariant the resulting graph, say T_γ also C_γ -invariant and we take the quotient $\bar{T}_\gamma$. This $\bar{T}_\gamma$ is not always a tree. However, if we remove the (finite!) subgraph spanned by those $V \subset \Gamma$ on which the function d_γ is $\leq 200(|\gamma| + \delta + 1)$ then what remains is a union of finitely many trees. One sees as earlier that these trees are Markov and the proof is concluded.

8.5.P One can reformulate Theorem 8.5.N as follows.

The counting function for the intersections of the set $\Gamma\gamma\Gamma^{-1}$ with concentric spheres in Γ is rational for all $\gamma \in \Gamma$.

Since Γ contains only finitely many conjugacy classes of torsion elements, we see with the above that the counting function for torsion (as well as for non-torsion) elements in Γ is rational (compare 5.2).

8.5.Q Let X be the Cayley graph of a word hyperbolic group Γ and $\pi: \Phi \rightarrow \partial X = \partial \Gamma$ be the Markov presentation over Γ provided by 8.4. Recall that points in Φ are represented by certain (distance like) functions $\varphi: \Gamma \rightarrow \mathbb{Z}$ defined up to additive constant. This presentation of $\partial \Gamma$ is closely related to the Markov chain (over $\mathbb{Z}_+$) associated to the tree $\tilde{T}$. Namely, let $\gamma = \gamma(\varphi) \in \Gamma$ be the first (for a fixed ordering) generator γ of Γ , such that $\varphi(\gamma) = \varphi(\text{id}) - 1$, and let us define $\alpha: \Phi \rightarrow \Phi$ by $\gamma^{-1}\varphi$ for all $\varphi \in \Phi$ and $\gamma = \gamma(\varphi)$. Clearly, this α is identical to the (one-sided) Markov shift associated to $\tilde{T}$. Then we complete α to a two-sided Markov shift $\bar{\alpha}$ on the space $\bar{\Phi}$ of $\mathbb{Z}$ -orbits of α that are maps $\psi: \mathbb{Z} \rightarrow \bar{\Phi}$ satisfying $\psi \circ s = \alpha \circ \psi$ for the translation $s: i \mapsto i + 1$ on $\mathbb{Z}$. Let us give a description of $\bar{\Phi}$ as of a Γ -quotient of the following space $\tilde{\Phi}$. Each $\tilde{\varphi} \in \tilde{\Phi}$ is given by a (distance-like) function $\varphi: X \rightarrow \mathbb{R}$ and a (gradient) line $\ell: \mathbb{R} \rightarrow X$, such that

- (a) $\ell(\mathbb{Z})$ lies in the zero skeleton of X
 - (b) for every $i \in \mathbb{Z}$ the vertex $\ell(i+1)$ is the first vertex adjacent to $\ell(i)$ such that $\varphi(\ell(i+1)) = \varphi(\ell(i)) - 1$, where "the first" refers to the order on edges corresponding to that on the generators of Γ .
- Recall that function φ is quasiconvex and distance-like:

$$\varphi(x) = \varphi(y) + \text{dist}(x, \varphi^{-1}(-\infty, y])$$

for all x and y in X satisfying $\varphi(x) \geq \varphi(y)$. Hence, the values of φ on 20δ -neighborhood of $\ell(\mathbb{R}) \subset X$ determine φ on all of X . Now, Γ obviously acts on $\tilde{\Phi}$ and also $\mathbb{Z}$ acts on $\tilde{\Phi}$ by $\bar{\alpha}: (\varphi, \ell) \mapsto (\varphi, \ell \circ s)$ for the translation s on $\mathbb{Z}$. Clearly

$$(\tilde{\Phi}, \bar{\alpha})/\Gamma = (\bar{\Phi}, \bar{\alpha}).$$

Let us relate $\tilde{\Phi}$ to the space $\hat{G} = \hat{G}(\Gamma)$ of geodesics in X . Denote by $\bar{\Phi}_{\mathbb{R}}$ the mapping torus of $\bar{\alpha}$. That is

$$\bar{\Phi}_{\mathbb{R}} = (\bar{\Phi} \times \mathbb{R})/\mathbb{Z},$$

where $\mathbb{Z}$ acts by $\bar{\alpha}$ on $\bar{\Phi}$ and by the unit shift on $\mathbb{R}$. There is an (obviously defined) action of $\mathbb{R}$ on $\bar{\Phi}_{\mathbb{R}}$, called $\bar{\alpha}_t: \bar{\Phi}_{\mathbb{R}} \rightarrow \bar{\Phi}_{\mathbb{R}}$ for $t \in \mathbb{R}$, such that the map $\bar{\alpha}_1$ sends $\bar{\Phi} = \bar{\Phi} \times 0 \subset \bar{\Phi}_{\mathbb{R}}$ onto itself and $\bar{\alpha}_1|_{\bar{\Phi}} = \bar{\alpha}$. One sees as earlier that $\bar{\Phi}_{\mathbb{R}} = \tilde{\Phi}_{\mathbb{R}}/\Gamma$ where $\tilde{\Phi}_{\mathbb{R}}$ consists of pairs (φ, ℓ) , where the above condition (a) on ℓ is not required any more and (b) applies to the pairs $(\varphi(t), \varphi(t+1))$, such that $\varphi(t)$ lies in the zero skeleton of X .

Now the projection $(\varphi, \ell) \mapsto \ell$ maps $\tilde{\Phi}_{\mathbb{R}}$ into the space $G(X)$ of all lines in X . Then we recall the (non-unique) map of $G(X)$ onto $\hat{G} = \hat{G}(\Gamma)$ and we denote the composed map by $\tilde{\pi}_{\mathbb{R}}: \tilde{\Phi}_{\mathbb{R}} \rightarrow \hat{G}$. Clearly, $\tilde{\pi}_{\mathbb{R}}$ is surjective and the (identification) relation, called

$$\tilde{R}_{\mathbb{R}} = \{(\tilde{\varphi}_1, \tilde{\varphi}_2) \mid \tilde{\pi}_{\mathbb{R}}(\tilde{\varphi}_1) = \tilde{\pi}_{\mathbb{R}}(\tilde{\varphi}_2)\} \subset \tilde{\Phi}_{\mathbb{R}} \times \tilde{\Phi}_{\mathbb{R}}$$

satisfies for all $\tilde{\varphi}_1 = (\varphi_1, \ell_1)$ and $\tilde{\varphi}_2 = (\varphi_2, \ell_2)$ in $\tilde{\Phi}_{\mathbb{R}}$:

$$(\tilde{\varphi}_1, \tilde{\varphi}_2) \in \tilde{R}_{\mathbb{R}} \Rightarrow |\ell_1(t) - \ell_2(t)| \leq d,$$

for all $t \in \mathbb{R}$, where $d = d(\Gamma)$ is some constant. Also notice that the map $\tilde{\pi}_{\mathbb{R}}$ is *injective* on every line in $\tilde{\Phi}_{\mathbb{R}}$ and every $\mathbb{R}$ -orbit $\tilde{\ell}$ of $\tilde{\Phi}_{\mathbb{R}}$ is homeomorphically mapped onto the unique $\mathbb{R}$ -orbit in $\hat{G}$ having some limit points in $\partial\Gamma$ as $\tilde{\ell}$. Thus the relation $\tilde{R}_{\mathbb{R}}$ is *strictly* contained in the (neighborhood) relation $|\ell_1(t) - \ell_2(t)| \leq d$, called $\tilde{R}(d) \subset \tilde{\Phi}_{\mathbb{R}} \times \tilde{\Phi}_{\mathbb{R}}$. In fact, one could define (non-uniquely) $\tilde{R}_{\mathbb{R}}$ (and thus $\hat{G}$) as a closed Γ -invariant subset $\tilde{R}$ in $\tilde{R}(d)$ for some (sufficiently large) d , which is an *equivalence* relation, such that

(a) No two distinct points on same $\mathbb{R}$ -orbit in $\tilde{\Phi}_{\mathbb{R}}$ are $\tilde{R}$ -equivalent.

(b) If two $\mathbb{R}$ -orbits have same ends in $\partial\Gamma$, then $\tilde{R}$ establishes a homeomorphism between these orbits.

(The existence of such an $\tilde{R}$ is proven in 8.3.)

Also notice that since the presentation $\pi: \Phi \rightarrow \partial\Gamma$ is finite-to-one, say at most N -to-one, the (quotient) map $\tilde{\pi}_R: \tilde{\Phi}_R \rightarrow \hat{G}$ is at most M -to-one for some M satisfying $M \leq Nk^d$, where k is the number of generators in Γ .

Now we observe that the intersection $\tilde{R}(d) \cap \tilde{\Phi} \times \tilde{\Phi}$ is univariant under the $\mathbb{Z}$ - and Γ -actions and thus it projects to a certain subset (relation) in $\bar{\Phi} \times \bar{\Phi}$, called $\bar{R}(d) \subset \bar{\Phi} \times \bar{\Phi}$ which is invariant under the (diagonal) action of $\mathbb{Z}$. By looking through the definitions, one easily sees the following

8.5.R Properties of $\bar{R}(d)$: The subset $\bar{R}(d)$ is Markov; it is symmetric under the involution $(\bar{\varphi}_1, \bar{\varphi}_2) \leftrightarrow (\bar{\varphi}_2, \bar{\varphi}_1)$ but (in general) it is not an equivalence relation. However, there exists an integer $i_0 > 0$, such that if $(\bar{\varphi}_1, \bar{\varphi}_2) \in \bar{R}(d)$ and $(\bar{\varphi}_2, \bar{\varphi}_3) \in \bar{R}(d)$ then $(\bar{\varphi}_1, \bar{a}^i \bar{\varphi}_3) \in \bar{R}(d)$ for some $i \in \mathbb{Z}$ satisfying $|i| \leq i_0$.

Finally, the projection of $\bar{R}(d)$ to Φ is at most M_d -to-one for some constant $M_d < \infty$.

Remark: The weak transitivity of $\bar{R}(d)$ stated above shows that $\bar{R}(d)$ induces an equivalence relation on the set of $\mathbb{Z}$ -orbits in $\bar{\Phi}$. In fact this equivalence relation identifies two orbits if and only if they go to same orbit under the map $\bar{\Phi} \rightarrow \bar{G} \stackrel{\text{def}}{=} \hat{G}/\Gamma$ obtained by comparing the inclusion $\bar{\Phi} \subset \tilde{\Phi}_R$ and the (quotient) map $\tilde{\Phi}_R \rightarrow \bar{G}$. Notice that the map $\bar{\Phi} \times \bar{G}$ induces a surjective map on the sets of orbits (it sends $\mathbb{Z}$ -orbits of $\bar{\Phi}$ to $\mathbb{R}$ -orbits of $\bar{G}$), such that the set of periodic orbits in $\bar{\Phi}$ bijectively goes onto the set of periodic orbits in $\bar{G}$.

8.5.S Let us explain the algebraic meaning of periodic orbits in $\bar{G}$. Call γ_1 and γ_2 in Γ stably conjugate if γ_1^m is conjugate to γ_2^m for some integer m . Denote the stable conjugacy class of γ by $[\gamma]_{\text{st}}$, (observe that it contains the conjugacy class $[\gamma]$ of γ) and define the stable exponent

$$p[\gamma]_{st} = \lim_{i \rightarrow \infty} \inf i^{-1} p(\gamma^i)$$

where $p(\gamma)$ denote the exponent of γ that is the maximal integer p such that $\gamma = \gamma_0^p$ for some $\gamma_0 \in \Gamma$.

Now we recall the maximal elementary (i.e. isomorphic to a finite extension of $\mathbb{Z}$) subgroup $E_\gamma \ni \gamma$ in Γ attached to every non-torsion $\gamma \in \Gamma$. Every stable conjugation between γ_1 and γ_2 also conjugate E_{γ_1} onto E_{γ_2} . It follows that the stable exponent is an integer. In fact, non-torsion elements γ_1 and γ_2 are stably conjugate if $p[\gamma_1]_{st} = p[\gamma_2]_{st}$ and if some $\gamma \in \Gamma$ acting on $\partial\Gamma$ sends $(\gamma_1^{-\infty}, \gamma_1^{\infty}) \mapsto (\gamma_2^{-\infty}, \gamma_2^{\infty})$.

Now we see that there is a one-to-one correspondence between stable conjugacy classes of non-torsion $\gamma \in \Gamma$ and periodic orbits in $\tilde{G}$, such that the stable exponents of γ equals the multiplicity of the corresponding orbits.

8.5.T Let us turn to periodic points (and orbits) of $(\tilde{\Phi}, \tilde{\alpha}) = (\tilde{\Phi}, \tilde{\alpha})/\Gamma$. Take a fixed point of some power of $\tilde{\alpha}$, say $\tilde{\varphi} \in \text{Fix } \tilde{\alpha}^k \subset \tilde{\Phi}$, and let $\tilde{\varphi} \in \tilde{\Phi}$ over $\tilde{\varphi}$ be represented by (φ, ℓ) , where ℓ is a line in X and φ is a (distance-like) function in the 20δ -neighborhood of $\ell \subset X$. Since $\tilde{\varphi}$ is fixed under π^k there exists a non-torsion $\gamma \in \Gamma$ with $\gamma^{\pm\infty} = \ell(\pm\infty)$, such that the line ℓ is invariant under γ and $\varphi \circ \gamma = \varphi - k$. It is clear that the stable conjugacy class of such a γ only depends on k and the (finite) $\tilde{\alpha}$ -orbit of $\tilde{\varphi}$ and that $|[\gamma]_-| = |[\gamma]| = k$, where $|[\gamma]_-| = \lim_{i \rightarrow \infty} \inf i^{-1} |\gamma^i|$ and $|[\gamma]| = \inf_{\gamma' \in [\gamma]} |\gamma'|$ (compare 5.2). Furthermore, since the correspondence between the orbits in $\tilde{\Phi}$ and $\hat{G}$ is finite-to-one, there exists an integer m_0 depending only on X , such that every non-torsion element in Γ whose stable exponent is divisible by m_0 is represented by some k and $\tilde{\varphi}$. This in particular, implies that the number $m_0 |[\gamma]_-|$ is an integer for all $\gamma \in \Gamma$.

The above relation between periodic orbits and classes $[\gamma]_{st}$ can be reformulated as follows. Recall the embedding

$\bar{\Phi} = \Phi \times 0 \subset \bar{\Phi}_{\mathbb{R}}$, let

$$\bar{\Phi}' = (\bar{\Phi} \times 0) \cup (\bar{\Phi} \times (\frac{1}{m_0})) \cup (\bar{\Phi} \times (\frac{2}{m_0})) \cup \dots \cup (\bar{\Phi} \times 0 = \bar{\Phi} \times 1)$$

and consider the transformation $\bar{\alpha}' = \bar{\alpha}_t$ for $t = 1/m_0$ on $\bar{\Phi}'$.

Notice that $(\bar{\alpha}')^{m_0} \mid \Phi \times 0 = \bar{\alpha}$ and that $\bar{\Phi}' = \tilde{\Phi}'/\Gamma$ where $\tilde{\Phi}'$ consists of the pairs (φ, ℓ) where $\ell(0) \in X$ is of the form,

$$\ell(0) = \frac{1}{m_0} x_0 + \frac{m_0 - i}{m_0} x_1,$$

or some integer i and two adjacent vertices in X .

Now one can obviously find (by no means uniquely) an $\bar{\alpha}'$ -invariant equivalence relation $\bar{R}' \subset \bar{\Phi} \times \bar{\Phi}'$, such that the periodic orbits of the corresponding quotient of $(\bar{\Phi}', \bar{\alpha}')$, say of $(\hat{\Phi}', \hat{\alpha}')$ are in a one-to-one correspondence with stable conjugacy classes in Γ . (Notice that we make no assumptions on the topology of $\bar{R}'$ and view $\hat{\Phi}'$ as an abstract set.) Namely, the pairs $(k, \hat{\psi})$, where $k = 1, 2, \dots$, and $\hat{\psi} \subset \hat{\Phi}'$ is a k -periodic orbit in $\bar{\Phi}'$, correspond to the classes $[\gamma]_{st}$, such that $|\gamma|_- = k/m_0$.

Next we consider the relation $\bar{R}'(d) \subset \bar{\Phi}' \times \bar{\Phi}' \subset \bar{\Phi}_{\mathbb{R}} \times \bar{\Phi}_{\mathbb{R}}$ which is the Γ -quotient of the $|\ell_1 - \ell_2| \leq d$ relation on $\tilde{\Phi}'$. Clearly, this $\bar{R}'(d)$ has the same properties as $\bar{R}(d)$ discussed earlier. In particular, $\bar{R}'(d)$ is Markov, the projections of $\bar{R}'(d)$ to $\bar{\Phi}'$ is finite-to-one and $\bar{R}'(d)$ (though not an equivalence relation) satisfies a weak transitivity property. It follows that one can find the above relation $\tilde{R}'$ such that $\bar{R}'(d) \supset \tilde{R}'$.

Let us summarize pertinent properties of $\bar{R}'$ in the following

Definition: Let Ψ be a Markov shift over $\mathbb{Z}$ and $R \subset \Psi \times \Psi$ be a $\mathbb{Z}$ -invariant equivalence relation. Call R *sub-Markov* if there exists a subshift $R_0 \subset \Psi \times \Psi$ containing R and an integer i_0 , such that the set of pairs $(\psi_1, i\psi_2)$ for all $(\psi_1, \psi_2) \in R$ and $i = 1, 2, \dots, i_0$, contains R_0 , where $i\psi$ denotes the action of $i \in \mathbb{Z}$ in $\psi \in \Psi$.

Now we can say that the relation $\bar{R}'$, for which the periodic orbits in $\bar{\Phi}'/\bar{R}'$ one-to-one correspond to stable conjugacy classes,

can be chosen sub-Markov.

8.5.U Theorem: Let R be a sub-Markov equivalence relation on Ψ and let $\pi(k)$ denote the number of k -periodic (i.e. fixed for $k\mathbb{Z} \subset \mathbb{Z}$) points in Ψ/R . If the (quotient) map $\pi: \Psi \rightarrow \Psi/R$ is at most N -to-one for some $N < \infty$, then the counting function $Z(t) = \sum_k \pi(k)t^k$ is rational.

Remarks:

(a) This theorem for R Markov is due to Manning (see [Ma]) who, in fact proves that the corresponding ζ -function $\exp \sum_k k^{-1} \pi(k)t^k$ is rational. Our argument below is similar to that of Manning and implies the rationality of the ζ -function.

(b) It is unclear if the "finite-to-one" assumption is essential.*

(c) If Γ has no torsion, then the counting function for the action on $\bar{\Phi}'/\bar{R}'$ is identical to that in 5.2.D. Thus Theorem 5.2.D follows from 8.5 and the preceding discussion.

8.5.V The proof of 8.5.U: Start with the following simple

Lemma: Let (Φ, α) be a Markov shift over $\mathbb{Z}$ and let $\beta: \Phi \rightarrow \Phi$ be a continuous map of finite order commuting with α . Then the counting function $Z(t) = Z(t; \alpha, \beta)$ defined by

$$Z(t) = \sum_k (\#\{\varphi \in \Phi \mid \alpha^k \varphi = \beta \varphi\}) t^k$$

*The "finite to one" condition is redundant for finitely presented $\mathbb{Z}$ -systems, see [Fr].

is rational.

Proof: One can assume without loss of generality that (Φ, α) is represented by some zero-one (Markov) matrix $M = M(i, j)$ for $1 \leq i, j \leq n$, such that the action of β is given by some permutation β' on the index set $\{1, 2, \dots, n\}$. Then $Z(t) = \sum_k t^k \sum_{i=1}^n M^k(i, \beta'(i))$ and the proof follows.

Now we prove the theorem by breaking our original $Z(t)$ into a finite linear combination of the function $Z(t, \alpha_\nu, \beta_\nu)$ for some auxiliary Markov shifts $(\Phi_\nu, \alpha_\nu, \beta_\nu)$. These shifts will be subshifts in the Cartesian products Ψ^m for $m = 1, 2, \dots, n$ for some fixed n depending on N and i_0 . Namely we consider in Ψ^m all possible intersections of the subshifts of the following two types

$$R = R(i_1, j_1; i_2, j_2) = \{ \psi_1, \dots, \psi_m \mid (i_1 \psi_{j_1}, i_2 \psi_{j_2}) \in R_0 \}$$

and

$$\Delta = \Delta(i_1, j_1; i_2, j_2) = \{ \psi_1, \dots, \psi_m \mid (i_1 \psi_{j_1} = i_2 \psi_{j_2}) \}$$

where $1 \leq j_1, j_2 \leq m$ and $-i_m \leq i_1, i_2 \leq i_m$ for some constant i_m depending on N and i_0 .

Some among these intersections $\Phi_\nu \subset \Psi^m$ are invariant under certain permutation β_ν (of the index set $\{1, \dots, m\}$) acting on Φ^m and thus we get all our $(\Phi_\nu, \alpha_\nu, \beta_\nu)$ where α_ν stands for the (diagonal) action of the original shift on $\Phi_\nu \subset \Psi^m$. Besides the Z -functions corresponding to these Φ_ν our $Z(t)$ also will contain finitely many terms of the form $\sum_k t^{ik}$ corresponding to i -periodic points with $i \leq i(i_0, N)$.

The counting process is naturally divided into two steps. The first can be expressed by the following trivial lemma.

Consider an arbitrary action of $\mathbb{Z}$ on a set Ψ with finitely many points of every fixed period and let $R \subset \Psi \times \Psi$ be a $\mathbb{Z}$ -invariant equivalence relation, such that the projections $R \rightarrow \Psi$ is at most N -to-one for $N < \infty$. Denote by $R_m \subset \underbrace{\Psi \times \Psi \times \dots \times \Psi}_m$

the subset,

$$R_m = \{ \psi_1, \dots, \psi_m \mid (\psi_i, \psi_j) \in R, 1 \leq i, j \leq m \}.$$

Notice that R_m naturally acted upon by $\mathbb{Z}$ and by the permutation group S_m .

Lemma: *Let (counting) Z-function for Ψ/R is a universal rational linear combination of Z-functions for (R_m, σ) for finitely many (depending on N) $n = 1, 2, \dots$ and $\sigma \in S_m$.*

In order to conclude the proof of 8.5.U we must relate σ -periodic points of R_m^n to those of the (Markov!) relation $(R_0)_m$. Since $R_m \subset (R_0)_m$ every σ -periodic point in R_m corresponds to one or more (but finitely many) σ -periodic points in $(R_0)_m$. To count the multiplicity we project $(R_0)_m$ onto R_m by

$$(\psi_1, \dots, \psi_m) \mapsto (i_1 \psi_1, \dots, i_m \psi_m)$$

where $(i_1, \dots, i_m)$ is the first (for the lexicographic order) n -tuple of integers i_j with $|i_m| \leq i_m$ which is contained in R_m . Though this projection does not commute with permutations $\sigma \in S_m$, one can obviously express the counting function for σ -periodic points in R_m in terms of similar functions for $R'_m \subset R_m^{m'} \subset \Phi^{mm'}$ where $R' \subset (R_0)_m \times (R_0)_m$ is the (Markov!) equivalence relation for the projection $(R_0)_m \rightarrow R_m$, provided there is no point in Ψ with period $\leq k_0$ for some constant $k_0 = k_0(N, i_0)$. Since these (infinitely many) short period points contribute a rational summand to our Z-functions, the theorem follows from the lemmas.

8.5.W The proof of 5.2.D: Theorem 8.5.U implies the rationality of the counting function for *stable* conjugacy classes in Γ . Moreover, it implies the following more general fact. Let $\Gamma_0 \subset \Gamma$ be a quasiconvex subgroup and set

$$[N_k^0]_{st} = \sum p^{-1}(\gamma) |[\gamma]|_-,$$

where the sum is taken over the stable conjugacy classes of non-torsion elements in Γ_0 whose stable norm (measured for the given word metric in Γ) satisfies $|[\gamma]|_- = k/m_0$.

A simple generalization of the earlier argument leads to the following

8.5.X Theorem: *The (counting) function*

$$Z(\Gamma_0, t) = \sum_k [N_k^0]_{st} t^k \quad (\text{as well as})$$

$$\xi(\Gamma_0, t) = \exp \sum_k k^{-1} [N_k^0]_{st} t^k \text{ is rational.}$$

Now one easily sees that the Z-function of 5.2.D is a linear combination of $Z(\Gamma_i, t)$, where Γ_i are centralizers of (finitely many) finite subgroups in Γ which are pairwise non-conjugate. Thus the proof of 5.2.D is concluded.

Remarks and open questions: It would be interesting to understand the Z-function with the (unstable) norm $|[\gamma]|$ instead of $|[\gamma]|_-$. Notice that $e_i(\gamma) = i|[\gamma]|_- - |[\gamma^i]|$ is eventually periodic in i for all $\gamma \in \Gamma$. Namely, there exist an integer i_0 depending only on Γ and the generating subset, such that $|e_i(\gamma)| \leq i_0$ and $e_i(\gamma) = e_{i+i_0}(\gamma)$ for $i \geq i_0$. This suggests a similar

periodicity in i for the (rational) functions $\sum_r N_{\gamma^i}(r) t^r$ (see 5.2) and this raises the counting problem for the conjugacy classes $[\gamma]$ according to the "type" of $\sum_r N_{\gamma}(r) t^r$ rather than $|[\gamma]|_-$.

Finally, one wishes to use these Z-functions to produce some invariants of Γ (independent of the word metric). One possible approach consists in a study of some generalization of Z-function for non-word metrics on Γ and of continuous families of those. Then one may try to control the modification of such a Z-function as one continuously deforms one word metric to another. A specific invariant of Γ (which, in fact, is a quasi-isometry invariant) is the dimension of the L_p -cohomology of Γ . This dimension is either zero or $+\infty$, but